NANDHINI S S

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PROFESSIONAL SUMMARY

Aspiring Data Scientist and AI & Data Science undergraduate with strong foundations in machine learning, deep learning, and data analytics. Experienced in data preprocessing, feature engineering, model development, evaluation, and deployment. Skilled in Python, SQL, Flask, recommendation systems, and building AI-driven applications. Passionate about transforming data into actionable insights through scalable and intelligent systems.

EDUCATION

- 2023-2027 B.Tech Artificial Intelligence and Data Science
  CGPA : 8.86
- 2022-2023 Class XII with 95% Ponjesly Public Matric Higher Secondary School

TECHNICAL SKILLS

Programming Languages : Python, SQL.
Data Analysis : Numpy, Pandas, Exploratory Data Analysis (EDA), Data Cleaning, Data Preprocessing.
Machine Learning : Model development & evaluation, supervised & unsupervised learning, recommendation systems, feature engineering, cross-validation, hyperparameter tuning.
Deep Learning : Neural Networks, TensorFlow, Keras.
Frameworks & Tools : Excel, PowerBI, Apache Kafka, Git, GitHub, MySQL, Statistics.
Modeling Techniques : SVD++, TF-IDF, Cosine Similarity, Clustering (KMeans), RFM Analysis

WORK EXPERIENCE

Data Science Intern – AK Infopark Pvt. Ltd

- Developed a hybrid recommendation system using SVD++, TF-IDF, and AI strategies (RFM, popularity, price, location).
- Built and deployed a Flask-based ML app with preprocessing, feature engineering, and model evaluation.

PROJECTS

1. Product Recommendation System

- Developed a hybrid recommendation system using SVD++ (collaborative filtering) and TF-IDF (content-based filtering) with AI strategies like RFM analysis, trending models, location-based personalization, and price sensitivity modeling. Tech Stack: Python, Scikit-learn, Pandas, Flask, SQL, SVD++, TF-IDF
- Built and deployed a full-stack Flask application handling data preprocessing, feature engineering, and model evaluation.

2. Smart Agriculture System

- Built an intelligent irrigation prediction system using IoT sensor data and machine learning models (Decision Tree, Logistic Regression) to enable data-driven farming decisions. Tech Stack: Python, Scikit-learn, Pandas, NodeMCU, IoT Sensors
- Achieved 98% accuracy through data preprocessing, feature engineering, and model optimization, identifying soil moisture as the key factor for optimized water usage.

3. AI Disease Prediction

- Developed a Flask-based disease prediction application using a Decision Tree Classifier trained on 100+ diseases with dynamic symptom selection and ML-based prediction workflows. Tech Stack: Python, Flask, Pandas, Scikit-learn, HTML, CSS
- Designed a responsive user interface providing real-time predictions to enhance user experience and interaction.

LANGUAGES

Tamil, English